Md (Muhammad) Shafiuzzaman

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Ph.D. candidate at the University of California, Santa Barbara, working at the intersection of **LLM/agentic AI**, **program analysis**, and **formal methods** to build practical security and software engineering tools. I develop scalable analysis engines, such as STASE, which has uncovered critical vulnerabilities in UEFI firmware and Linux kernels and is currently transitioning into U.S. Army adversarial-testing workflows. My research leverages **LLM-driven orchestration** and **neuro-symbolic reasoning** to enable automated, evidence-bearing security and correctness auditing through large-scale parallel analysis workflows for mission-critical software systems. Building on STASE, my SAILOR framework extends this approach to arbitrary C/C++ via LLM-orchestrated symbolic execution with sanitizer-backed validation, and has uncovered 30+ previously unknown memory-safety vulnerabilities in widely used open-source software.

Education

University of California, Santa Barbara (UCSB) Expected Spring 2027
Ph.D. Candidate, Computer Science
Advisor: Prof. Tevfik Bultan
Dissertation: *Scaling Symbolic Execution from Vulnerability Detection to Verified Repair*

Abstract: My dissertation addresses the challenge of scaling and automating formal verification for large-scale production-grade software systems. I integrate hybrid analysis with agentic AI to autonomously synthesize verification harnesses and precise signatures, enabling security auditing at the scale required by real-world adversarial workflows. This work bridges the gap between formal rigor and operational security, providing a neuro-symbolic foundation for trustworthy AI-enabled infrastructure.

University of Dhaka 2016
M.S., Software Engineering
Thesis: *A Proactive Self-Adaptive Framework for Context-Aware Mobile Applications*

University of Dhaka 2014
B.S., Software Engineering

Research Interests

- **Agentic AI for Software Engineering:** Designing neuro-symbolic agents and LLM-driven orchestration to automate complex development, testing, and maintenance tasks.
- **Trustworthy and Verifiable GenAI:** Building pipelines that translate LLM outputs into machine-checkable and evidence-bearing artifacts with formal security and correctness guarantees.
- **Formal Methods for Large-Scale Systems:** Leveraging High-Performance Computing (HPC) to perform deep semantic analysis on complex, production-grade codebases.
- **Autonomous Cyber Defense:** Closing the loop between automated vulnerability discovery, multi-step exploit-chain analysis, verified patch synthesis, and runtime enforcement.

Academic Research Experience

Graduate Student Researcher, Verification Lab (VLab), UCSB 2022–Present

- **Project: HARDEN (DARPA, U.S. Department of Defense)** 2022–Present
  - Lead Graduate Researcher for a three-phase, 48-month program aimed at securing the "root of trust" in complex, integrated computing systems.
  - Designed and built STASE, a hybrid static-symbolic engine to detect emergent execution vulnerabilities in UEFI firmware and Linux kernels.
  - Developed CHAINER for exploit-chain reasoning, utilizing memory-aware symbolic analysis and signature composition to detect deep-link vulnerabilities.
  - Orchestrated the transition of reproducible artifacts to U.S. Army workflows; evaluators cited "high automation capability" for mission-critical auditing.
  - As the sole surviving bug-discovery (TA1) performer, carried the UCSB team through to HARDEN's final phase on the strength of STASE; the 48-month program concluded in 2026.
- **Project: INGOTS (DARPA, U.S. Department of Defense)** 2026–Present
  - Follow-on program transitioned from HARDEN, on automated synthesis, modeling, and analysis of exploit chains across complex systems; extending CHAINER to system-scale chain discovery.
- **Project: Sailor (LLM-Enhanced Formal Methods, Agentic AI)** 2025–Present
  - Built SAILOR, extending STASE to arbitrary C/C++ via LLM-orchestrated symbolic execution with sanitizer-backed validation; disclosed 30+ previously unknown memory-safety vulnerabilities (see Security Disclosures).
  - Leading development of neuro-symbolic agents that iteratively refine verification harnesses using feedback from symbolic execution and compiler sanitizers.
  - Integrating Retrieval Augmented Generation (RAG) to automate stub and assertion generation for system-scale codebases.
- **Project: Side-Channel Analysis in Crypto Libraries (Amazon Research)** 2024–Present
  - Adapted the STASE analysis methodology to identify and quantify information leakages and side-channel vulnerabilities in production-scale cryptographic libraries.
- **Project: Vehicle Security (U.S. Army TARDEC)** 2023–2025
  - Designed automated behavior-inference techniques to capture ECU behavior as state-machine models for CAN-based vehicle network security.

Publications

(⋆ denotes the three most important publications selected for this application)

Core Research: Conference Papers

⋆ [C1] **Md Shafiuzzaman**, Achintya Desai, Laboni Sarker, and Tevfik Bultan. "STASE: Static Analysis Guided Symbolic Execution for UEFI Vulnerability Signature Generation." *Proc. 39th IEEE/ACM Int. Conf. on Automated Software Engineering (ASE)*, 2024.
DOI: 10.1145/3691620.3695543 [Artifact/Code]

[C2] Achintya Desai, **Md Shafiuzzaman**, Wenbo Guo, and Tevfik Bultan. "PAGENT: Program Analysis Guided LLM Agent for Proof-of-Concept Generation." *Proc. 35th ACM SIGSOFT Int. Symp. on Software Testing and Analysis (ISSTA)*, 2026.
Artifacts Evaluated: Available & Functional.

[C3] Achintya Desai, Ruochen Dai, Yanju Chen, Ky Ho, Austin Kee, Sophie Bulatovic, **Md Shafiuzzaman**, Tevfik Bultan et al. "Stateful Behavior Inference and Runtime Enforcement for Vehicle Network Security." *Proc. 3rd USENIX Symposium on Vehicle Security and Privacy (VehicleSec)*, 2025. usenix.org/system/files/vehiclesec25-desai.pdf

⋆ [C4] Seemanta Saha, Laboni Sarker, **Md Shafiuzzaman**, Chaofan Shou, Albert Li, Ganesh Sankaran, and Tevfik Bultan. "Rare Path Guided Fuzzing." *Proc. 32nd ACM SIGSOFT Int. Symp. on Software Testing and Analysis (ISSTA)*, 2023.
DOI: 10.1145/3597926.3598136 [Artifact/Code]

[C5] H. Roy, **Md Shafiuzzaman**, and M. Samsuddoha. "SRCS: A New Proposed Counting Sort Algorithm based on Square Root Method." *22nd Int. Conf. on Computer and Information Technology (ICCIT)*, 2019.
DOI: 10.1109/ICCIT48885.2019.9038601

[C6] M. Sujon, **Md Shafiuzzaman**, M. M. Rahman, and M. R. Rahman. "Characterization and Localization of Performance-bugs using Naive Bayes Approach." *5th Int. Conf. on Informatics, Electronics and Vision (ICIEV)*, 2016.
DOI: 10.1109/ICIEV.2016.7760110

[C7] **Md Shafiuzzaman**, N. Nahar and M. R. Rahman. "A proactive approach for context-aware self-adaptive mobile applications to ensure Quality of Service." *18th Int. Conf. on Computer and Information Technology (ICCIT)*, 2015.
DOI: 10.1109/ICCITechn.2015.7488131

Workshop Papers

[W1] Vishva Arasan, Mahima Chaudhary, Wendy Contreras Martinez, **Md Shafiuzzaman**, Achintya Desai, and Tevfik Bultan. "Bug Composition: Triggering Multiple Bugs with Agentic Driver Generation." *Proc. 1st Int. Workshop on Agentic AI for Next-Generation Software Development (AgenticDev)*, 2026.
DOI: 10.1145/3843282.3844434 (with mentored ERSP undergraduates)

Under Review / In Submission

⋆ [S1] **Md Shafiuzzaman**, Achintya Desai, Wenbo Guo, and Tevfik Bultan. "SAILOR: Guiding Symbolic Execution for Memory Safety Vulnerability Detection." *Submitted to USENIX Security Symposium*, 2027.

[S2] Achintya Desai, **Md Shafiuzzaman**, and Tevfik Bultan. "CHAINER: Discovering Vulnerability Chains with Memory-Aware Symbolic Analysis." *Submitted to ACM Transactions on Software Engineering and Methodology (TOSEM)*. [Artifact/Code]

Journal Articles & Peer-Reviewed Publications

[J1] T. Islam, **Md Shafiuzzaman**, et al. "Nitrogen fertilizer recommendation for paddies through automating the leaf color chart (LCC)." *Int. Journal of Advanced Computer Science and Applications*, 2020.

[J2] M. A. Hossain, **Md Shafiuzzaman**, et al. "Implementation of VDI based Computer Laboratory in University Education System." *IJCSNS Int. J. Comput. Sci. Netw. Secur*, 2020.

[J3] R. R. Ema, **Md Shafiuzzaman**, et al. "Performance Improvement of Network Coding for Heterogeneous Data Items." *Int. Journal of Advanced Computer Science and Applications*, 2021.

[J4] **Md Shafiuzzaman**, A. S. Ami, and R. Rahman. "A Proactive Self-Adaptive Framework for Context-Aware Mobile Applications." *Int. Journal of Computer Applications*, 2018.

Security Disclosures (Selected)

Previously unknown memory-safety vulnerabilities I found and disclosed responsibly upstream (primarily with SAILOR*); every finding is confirmed by a concrete sanitizer crash. 30+ disclosed across 18 real-world C/C++ codebases.*

- **mruby:** integer-overflow out-of-bounds write (CWE-190/787), fixed by the maintainer. issue #6937
- **SELinux (checkpolicy):** double free (CWE-415), patch authored by us and merged upstream. commit fb255b9
- **koboldcpp:** stack-buffer overflow across 6 legacy model loaders (CWE-787), published advisory rated High. GHSA-qhvp-gj7g-rw26
- **whisper.cpp, bark.cpp, encodec.cpp, biogpt.cpp:** model-loader memory-safety bugs (CWE-125/787), fixes submitted and accepted upstream (whisper.cpp #3956, #3957; bark.cpp #220; encodec.cpp #57; biogpt.cpp #27).
- **Under coordinated disclosure:** additional findings in LIBOQS (post-quantum cryptography), tinyexr, hnswlib, usearch, Annoy, stable-diffusion.cpp, OpenFHE, CTranslate2, and the llama.cpp/ggml stack.

Grant Writing & Funding Contributions

Contributed to technical design, preliminary data generation, and narrative writing for awarded/submitted proposals:

- **DARPA HARDEN (Phase 3)** 2025
  - **Role:** Authored the transition plan for integrating STASE into U.S. Army workflows.
- **NSF Future CoRe/SHF** 2025
  - *Project:* Secure and Verifiable Code Generation via Specialized Agents.
  - **Role:** Designed the multi-agent orchestration framework for translating natural language requirements into verified code artifacts using agentic feedback loops.
- **Sony Research Award Program** 2025
  - *Project:* LLM-enhanced formal methods for embedded Linux security.
  - **Role:** Primary author for the methodology on utilizing Retrieval-Augmented Generation (RAG) to synthesize formal specifications from kernel source code.
- **DARPA ASEMA** 2025
  - *Project:* Assessing Security of Encrypted Messaging Applications.
  - **Role:** Conceptualized automated symbolic analysis pipelines for identifying complex logic flaws in encrypted communication protocols.
- **NSF SaTC Medium** 2024
  - *Project:* Side-channels: Proofs, Attacks, and Defenses.
  - **Role:** Co-authored technical sections on leveraging LLM-guided harness synthesis to detect timing and resource leakages in cryptographic implementations.
- **DARPA TRACTOR** 2024
  - *Project:* Translating C to Rust (TRACTOR).
  - **Role:** Developed workflows for using symbolic execution to verify semantic equivalence between legacy C code and synthesized Rust implementations.
- **Amazon Trusted Crypto** 2024
  - *Project:* Detecting and Quantifying Information Leakages in Crypto Libraries.
  - **Role:** Applied STASE methodologies to discover and quantify side-channel information leakages in production-scale cryptographic libraries.

Teaching Experience

Instructor of Record, University of California, Santa Barbara Summer 2022

- *Course:* **CS130A: Data Structures & Algorithms I**

- Held full responsibility for curriculum design, delivering lectures, and creating assessments for upper-division students.
- *Course Materials:* github.com/shafiuzzaman-md/CS130A_Summer2022

Lecturer, Jashore University of Science & Technology (JUST) 2018–2021

- Delivered core undergraduate courses including **Software Engineering**, **Mobile App Development**, and **Algorithm Analysis**.
- Developed project-based learning modules for mobile development, resulting in 20+ student-led application prototypes.

Instructor, Bangladesh Air Force Academy Spring 2019

- *Course:* **Programming in C and Python**
- Designed an intensive introductory programming track for aviation cadets, focusing on computational logic and systems-level efficiency.

Teaching Assistant, University of California, Santa Barbara 2021–2022

- Courses: *Fundamentals of Database Systems*, *Data Structures & Algorithms I & II*.
- Led weekly discussion sections and held office hours to provide technical guidance on complex system implementation projects.

Pedagogical Training

Summer Teaching Institute for Associates (STIA), UCSB 2022

Completed a competitive, 40-hour pedagogical program focused on evidence-based teaching practices, inclusive classroom management, and learning objective alignment.

Mentoring & Leadership

Undergraduate Research Mentor, Early Research Scholars Program (ERSP) 2023–Present

- **Cohort 2025 (5 students):** Mentored a team on *"LLM-Assisted Fuzzing and Symbolic Execution"*; the project produced a peer-reviewed workshop paper on agentic exploit-chain discovery ([W1] above).
- **Cohort 2023 (4 students):** Supervised a year-long project on *"Vulnerability Signature Generation"*.

Graduate Student Mentor, UCSB Graduate Scholars Program 2025–2026

- Appointed to mentor two incoming Ph.D. students, providing academic orientation and advising on laboratory integration and research workflows.

Industry Engineering Experience

Software Engineer Intern, Office of Research, UC Santa Barbara Summer 2022

- Developed and implemented robust role-based access control (RBAC) and comprehensive audit logging for the Central Authentication Service (CAS), securing sensitive university data.

Software Engineer, Mobil Jamuna Lubricants, Bangladesh 2016–2018

- Re-engineered the Lube Enterprise System data pipeline to enhance throughput and ensure data integrity for nationwide logistics.
- Awarded the **Highest Performance Rating (Top 1%)** for technical excellence and system reliability in 2017.

Junior Software Engineer, Wolters Kluwer, USA 2014–2016

- Modernized mission-critical financial compliance modules processing sensitive U.S. mortgage data, ensuring adherence to strict federal security standards.

Honors & Awards

UCSB Chancellor's Scholarship, University of California, Santa Barbara 2024
UCSB Academic Excellence Fellowship, University of California, Santa Barbara 2021
National Merit Scholarship, Government of Bangladesh (Top 0.25% of cohort)

Service & Applied Impact

Professional Service

- **External Reviewer / Sub-reviewer:** *ICSE*, *FSE*, *ISSTA*, *ASE*, *FASE* (Reviewed 22 papers) 2023–2025

Administrative & Community Leadership

- **Lead Developer, University Admissions Platform (JUST):** Orchestrated the in-house implementation of a scalable admissions system to replace an outsourced legacy framework.
- **Volunteer Developer, "HelpMe" Safety App:** Contributed to the Bangladesh Ministry of Women and Children Affairs initiative to build reactive safety tools for the public.

References

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